Information Security and Cryptography - AI 331

Name : Jeetu Mehra
Roll No : U24AI032
Date : 3/8/26

Lab Assignment 1

1. Understand the basics of Cryptography and explore how it works.

Cryptography is the science of securing information by converting it into an unreadable form called ciphertext, so that only authorized people can read it. It ensures Confidentiality, Integrity, Authentication and Non-repudiation of data.

It works using algorithms and keys.

- Encryption : Converts plain text into ciphertext using an algorithm and a key.
- Decryption : Converts ciphertext back to plain text using the key.
- Key : A secret value used in encryption and decryption.
- Algorithm : A set of rules used to perform encryption/decryption.

Two types of Cryptography are:

1. Symmetric Key Cryptography - Same key is used for both encryption and decryption. (e.g. AES, DES)
2. Asymmetric Key Cryptography - Different keys are used for encryption and decryption (Public Key and Private Key). (e.g. RSA)

Cryptography is used in real life for secure communication, online banking, passwords, digital signatures, and data protection.

2. Study the elements of Information Security (CIA Triad) and how are they related to Cryptography.

The CIA Triad consists of three main principles:

1. Confidentiality : Ensures that information is accessible only to authorized users. Cryptography provides this through encryption.
2. Integrity : Ensures that data is not altered or tampered with. Cryptography provides this using techniques like Hash functions and digital signatures.
3. Availability : Ensures that authorized users have access to information when needed. Cryptography helps in secure key management and protects systems from attacks that can disrupt availability.

Thus, Cryptography is the foundation that supports and strengthens all three principles of Information Security.